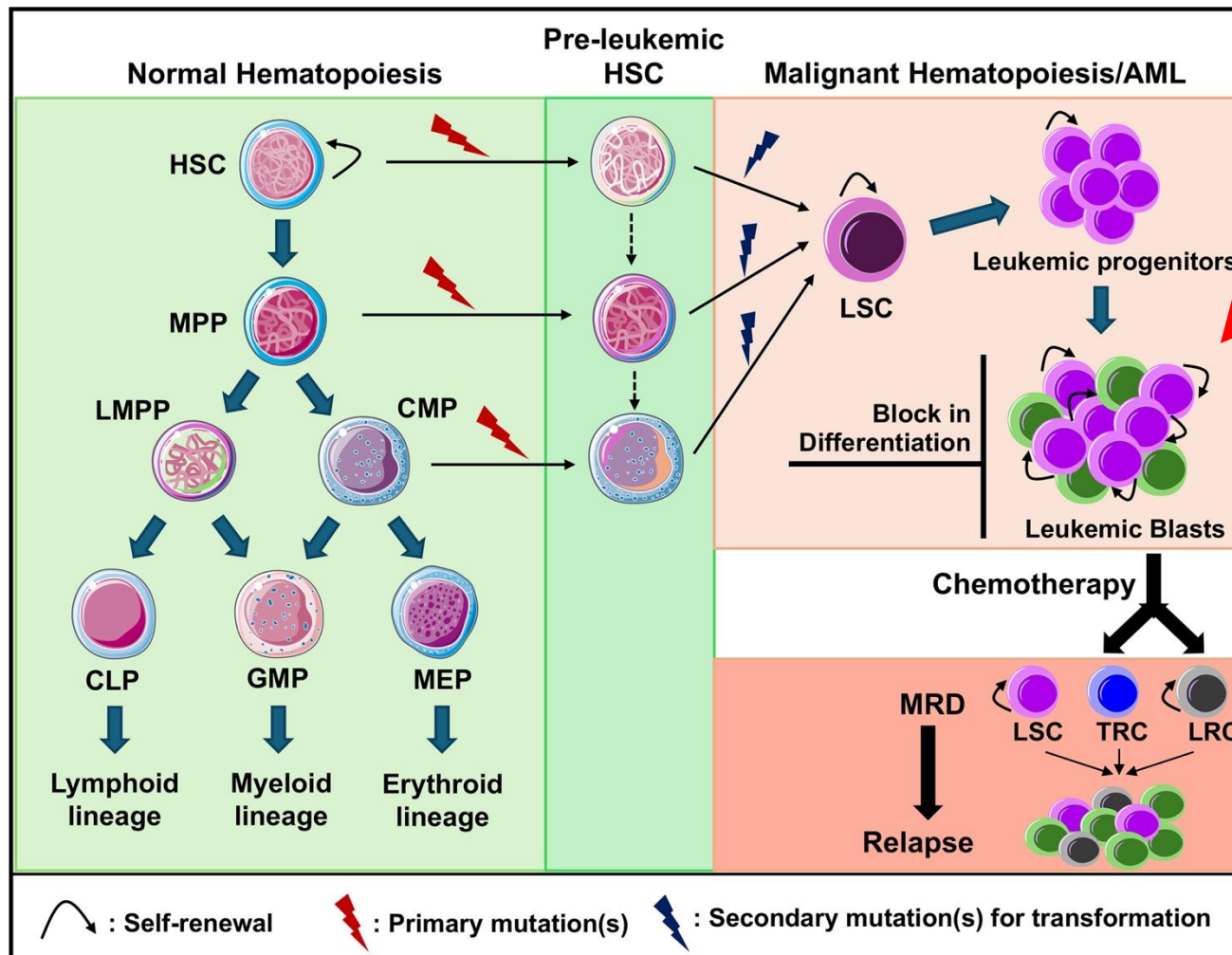


Dysregulated transcriptional programs in malignant cells of Acute Myeloid Leukemia (AML)

BCB420: Computational Systems Biology

Arhan Rupani



- Many malignant CellTypes/subclones within the same tumor, therefore, AML is a highly heterogenous disease
- Single cell sequencing gives us the ability to distinguish malignant from non-malignant cell types within a tumor, hence we can identify malignant subclones
- **Biological hypothesis:** we can compare malignant cells with its healthy counterpart to infer the mechanism of pathogenesis in AML

Dataset

Article



Single-Cell RNA-Seq Reveals AML Hierarchies Relevant to Disease Progression and Immunity

Peter van Galen,^{1,2,3,4,9} Volker Hovestadt,^{1,2,3,4,9} Marc H. Wadsworth II,^{3,5,6} Travis K. Hughes,^{3,5,6} Gabriel K. Griffin,^{1,2,3,4,7} Sofia Battaglia,^{1,2,3,4} Julia A. Verga,^{1,2,3,4} Jason Stephansky,^{2,7,8} Timothy J. Pastika,^{2,7,8} Jennifer Lombardi Story,⁴ Geraldine S. Pinkus,⁷ Olga Pozdnyakova,⁷ Ilene Galinsky,⁸ Richard M. Stone,⁸ Timothy A. Graubert,⁴ Alex K. Shalek,^{3,5,6} Jon C. Aster,^{2,7} Andrew A. Lane,^{2,8} and Bradley E. Bernstein^{1,2,3,4,10,*}

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¹⁰Lead Contact

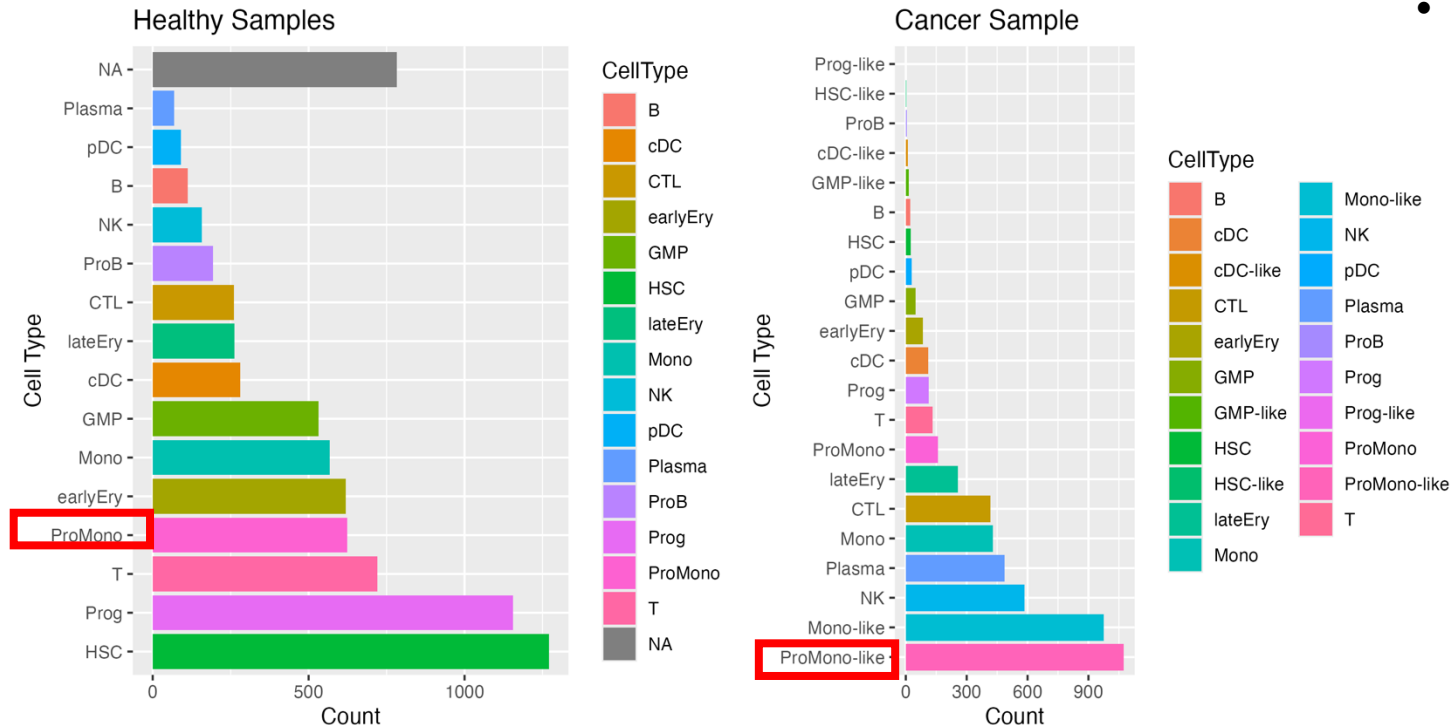
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- single cell RNAseq dataset to study AML by VanGalen et al.
- case control (tissue: bone marrow)
 - ~7000 cells (5 healthy donors) & ~30,000 cells (16 AML patients)

Downsampling strategy and method for data analysis

- Instead of analyzing all 15 patients at once, I analyze each patient at a time and then theoretically one can find a way to coalesce the results at the end
- Batch effects & Additional variability due to inter-tumor heterogeneity (Tirosh and Suva, 2024)
- Accordingly, the set up is:
 - 1 patient, ~5000 tumor cells vs ~7000 healthy cells => 12000 datapoints



- Insight from theory + data => there is a subclone of malignant promonocytes in this tumor

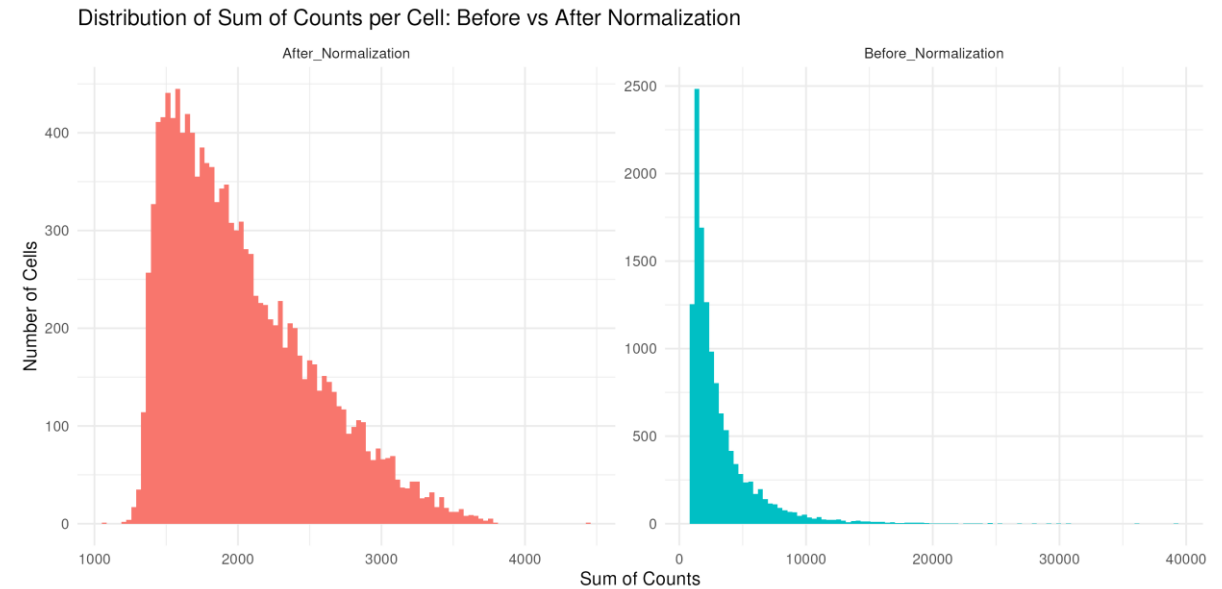
How does biological organization differ in malignant promonocytes from healthy promonocytes at the cellular and biomolecular level?

To answer this question, we will use the tools of systems biology learnt in BCB420!

How did I process this single cell dataset?

Pipeline:

- Quality control
 - For statistical robustness I removed cells with total expression levels 1.5 IQD away from the median
- Identifier mapping
 - Done by the authors
- Normalization by log transformation
 - For each cell, divide each gene's expression with total expression for that cell $\sim x$
 - $f(x) = \log(x(10000) + 1)$

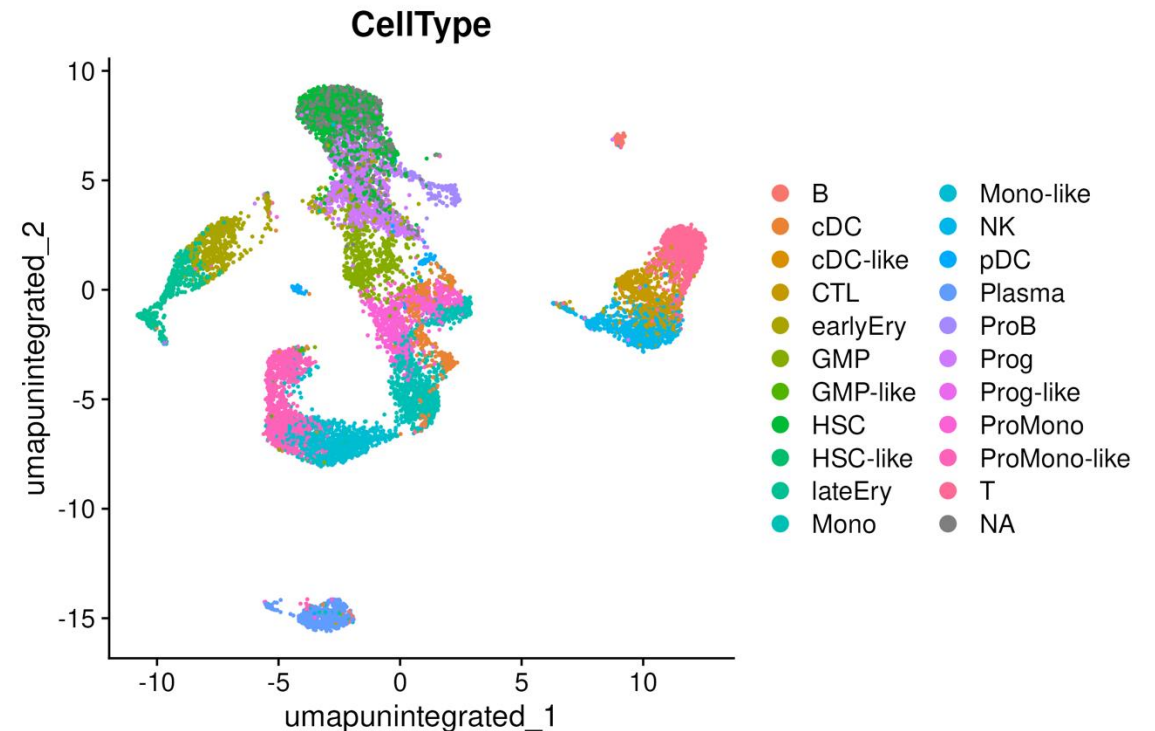


- Upon normalization, variability in the sum of counts decreases

How did I process this single cell dataset?

Pipeline:

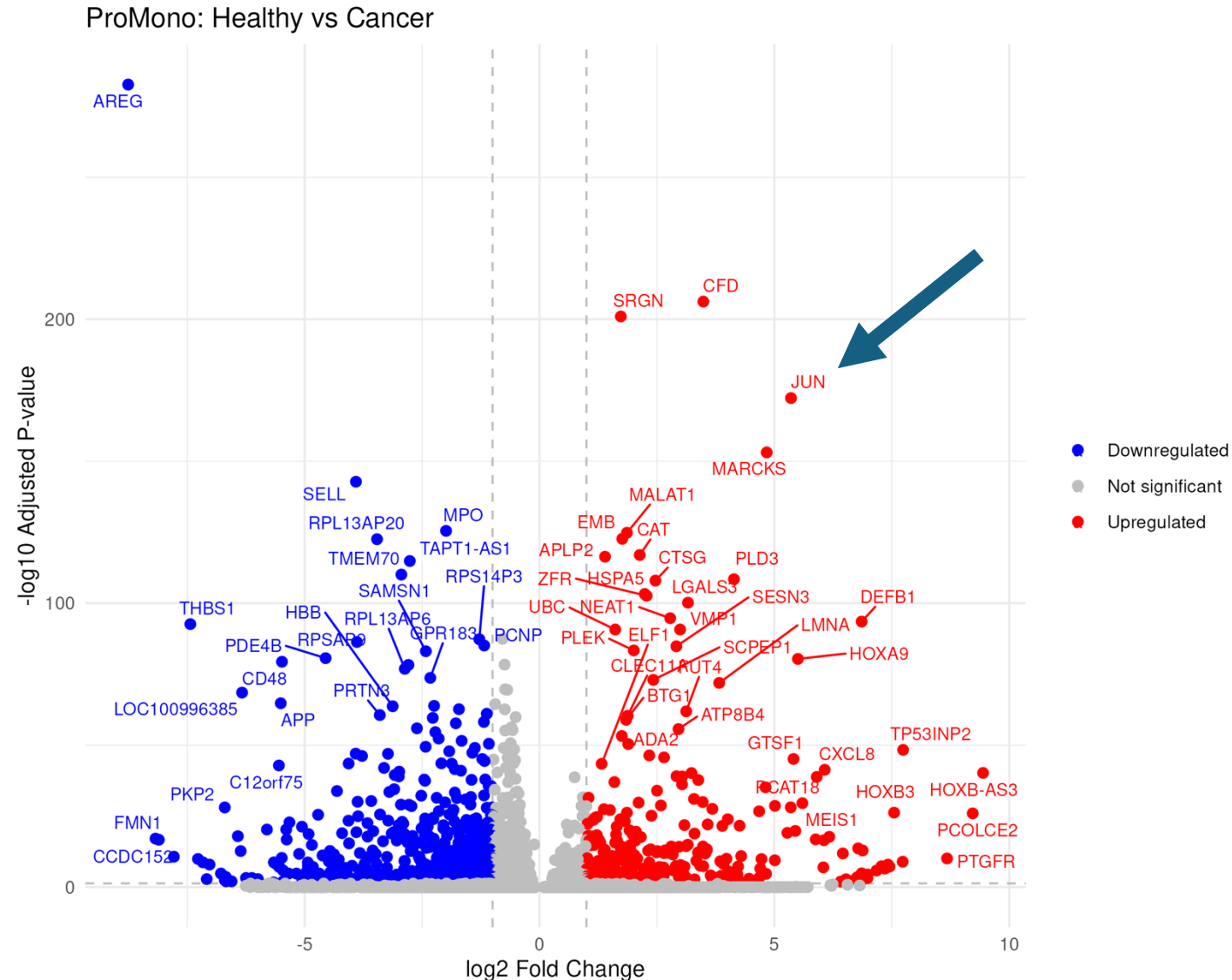
- Quality control
- Identifier mapping
- Normalization by log transformation
- Highly variable gene selection
 - Top 2000 genes with highest variance
- Dimensionality reduction (PCA)
 - Perform SVD on the expression matrix and keep eigenvectors with 50 highest eigenvalues
- Clustering (lieden) (not needed as we already have cell type annotations)
- Uniform Manifold Approximation (UMAP)



- UMAP visualization reveals a continuum of cell types typical of a developmental system

Downstream methods for biological interpretations from data

- Differential gene expression analysis between malignant vs healthy pro-monocytes
 - non-parametric Wilcoxon rank sum test
 - Adjusted p-value, based on Bonferroni correction
 - DEG cutoff: $\text{adj_pval} \leq 0.05$
- *JUN*: known protooncogene due to its involvement in proliferative signaling pathways



Downstream methods after data analysis

- Thresholded and non-thresholded gene set enrichment analysis
- non-thresholded analysis: I performed GSEA using the GUI of the original software
 - Baderlab geneset: Human GOBP; AllPathways; no PFOCR; no GO iea
 - 15 genes min; 500 genes max
 - 1000 permutations
- For the rest of the presentation I will focus on genesets which are significantly upregulated in malignant AML cells

GSEA Report for Dataset promono_healthy_vs_leukemia_expression

Enrichment in phenotype: na

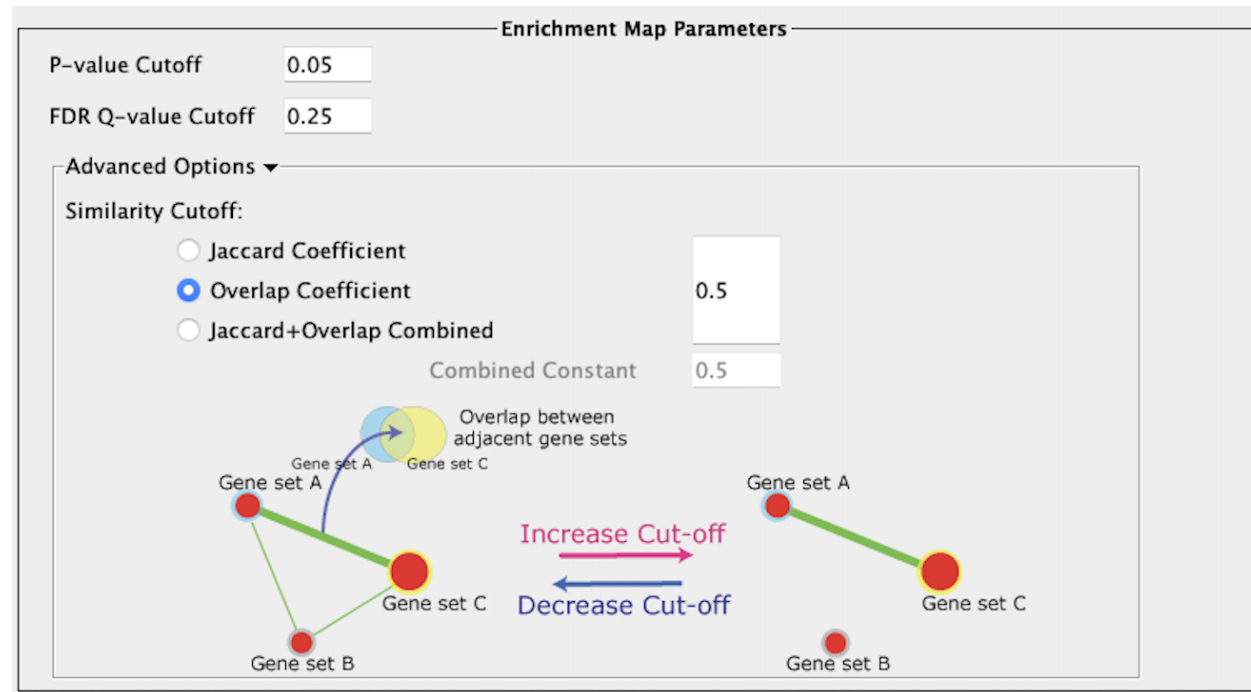
- 1633 / 4676 gene sets are upregulated in phenotype **na_pos**
- 25 gene sets are significant at FDR < 25%
- 64 gene sets are significantly enriched at nominal pvalue < 1%
- 222 gene sets are significantly enriched at nominal pvalue < 5%
- [Snapshot](#) of enrichment results
- Detailed [enrichment results in html](#) format
- Detailed [enrichment results in TSV](#) format (tab delimited text)
- [Guide to](#) interpret results

Enrichment in phenotype: na

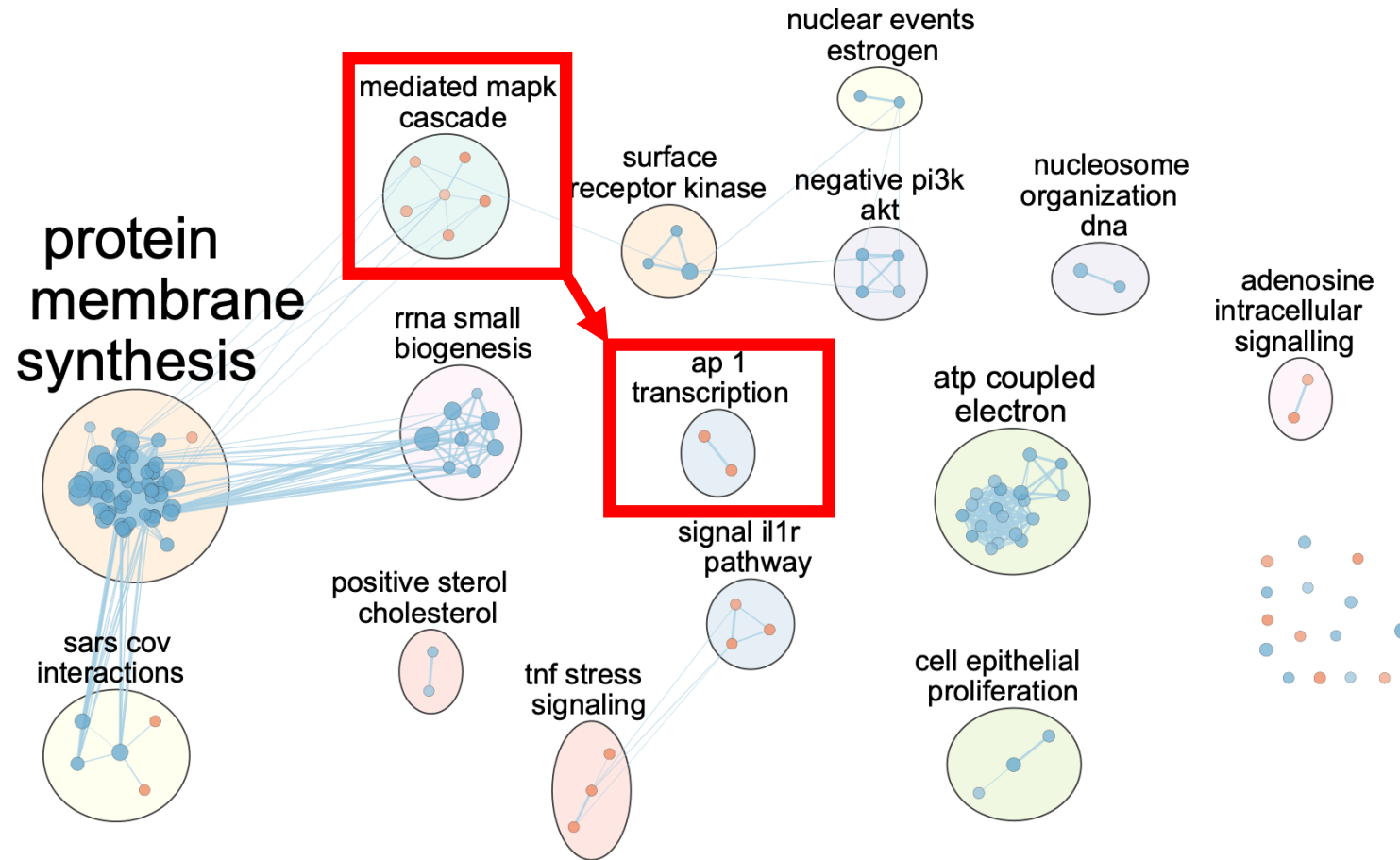
- 3043 / 4676 gene sets are upregulated in phenotype **na_neg**
- 126 gene sets are significant at FDR < 25%
- 144 gene sets are significantly enriched at nominal pvalue < 1%
- 275 gene sets are significantly enriched at nominal pvalue < 5%
- [Snapshot](#) of enrichment results
- Detailed [enrichment results in html](#) format
- Detailed [enrichment results in TSV](#) format (tab delimited text)
- [Guide to](#) interpret results

Cytoscape + Enrichment Map for network based visualization of GSEA Results

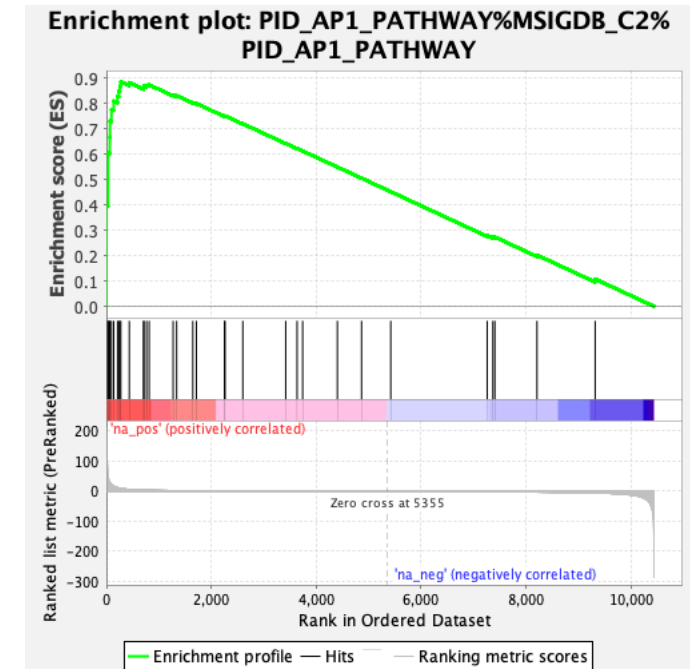
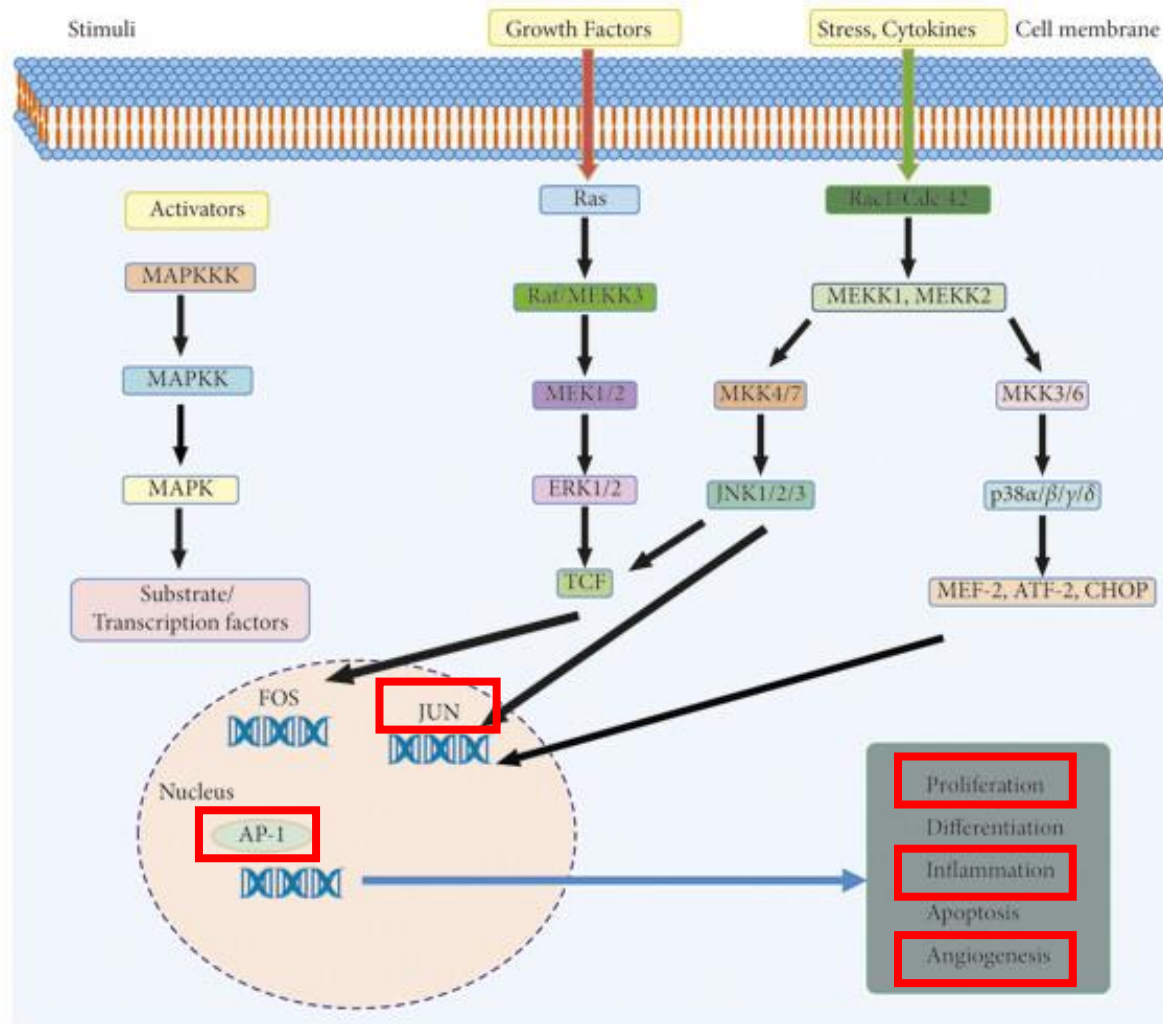
- Very permissive P-value and FDR thresholds with GSEA gene set permutation results (as a starting point)



Cytoscape + Enrichment Map for network based visualization of GSEA Results



Ap1 is a TF complex which has *JUN* as one of its subunits



- Similar results were obtained doing thresholded analysis with gprofiler using the same genesets for the upregulated gene list

Conclusions

- Overall, MAPK cascade and *AP1* TF are upregulated in malignant cells and could be causal to driving this patient's leukemia
- Genes encoding for the *AP1* TFs subunits like *JUN* and *FOS* could be potential therapeutic targets

Future Directions

- Run this pipeline on other patients and combine the results of GSEA from all analyses to identify recurring patterns in enriched gene sets and processes
- This can lead to novel pathways to target by precision medicine approaches specifically in AML
- This is much needed given the recalcitrant nature of this cancer

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Thank you!

